Euro-Shield 6Y Horizon

Structured Product Pricing

Case Study — ENERGETIC du CA

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Product Overview

- **Product Name:** Euro-Shield 6Y Horizon
- **Underlying:** DJ Euro Stoxx 50
- **Maturity (T):** 6 years
- **Spot (S_0):** 3,600 €
- **Guaranteed Capital:** 100%
- **Maximum Return:** 60%
- **Initial Nominal:** 100 €
- **Participation:** 100% (capped)

Step 1: Payoff Formalization

$$\text{Payoff}(S_T) = 100 \times \left[1 + \max\left(0, \min\left(0.60, \frac{S_T}{S_0} - 1\right)\right)\right]$$

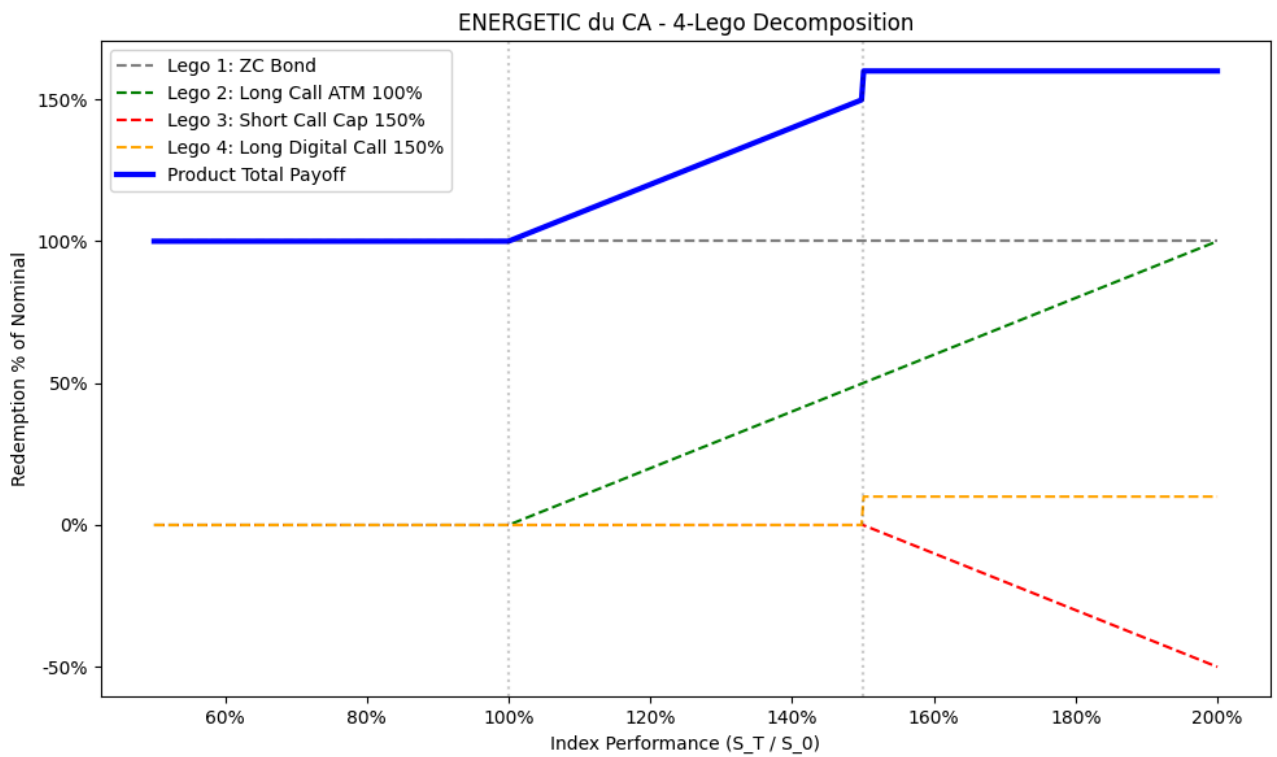
Scenario	Condition	Return
Bearish	$S_T \leq 3,600$	0%
Moderate Bull	$3,600 < S_T < 5,760$	0% - 60%
Strong Bull	$S_T \geq 5,760$	60% (cap)

Step 1: Lego Block Decomposition

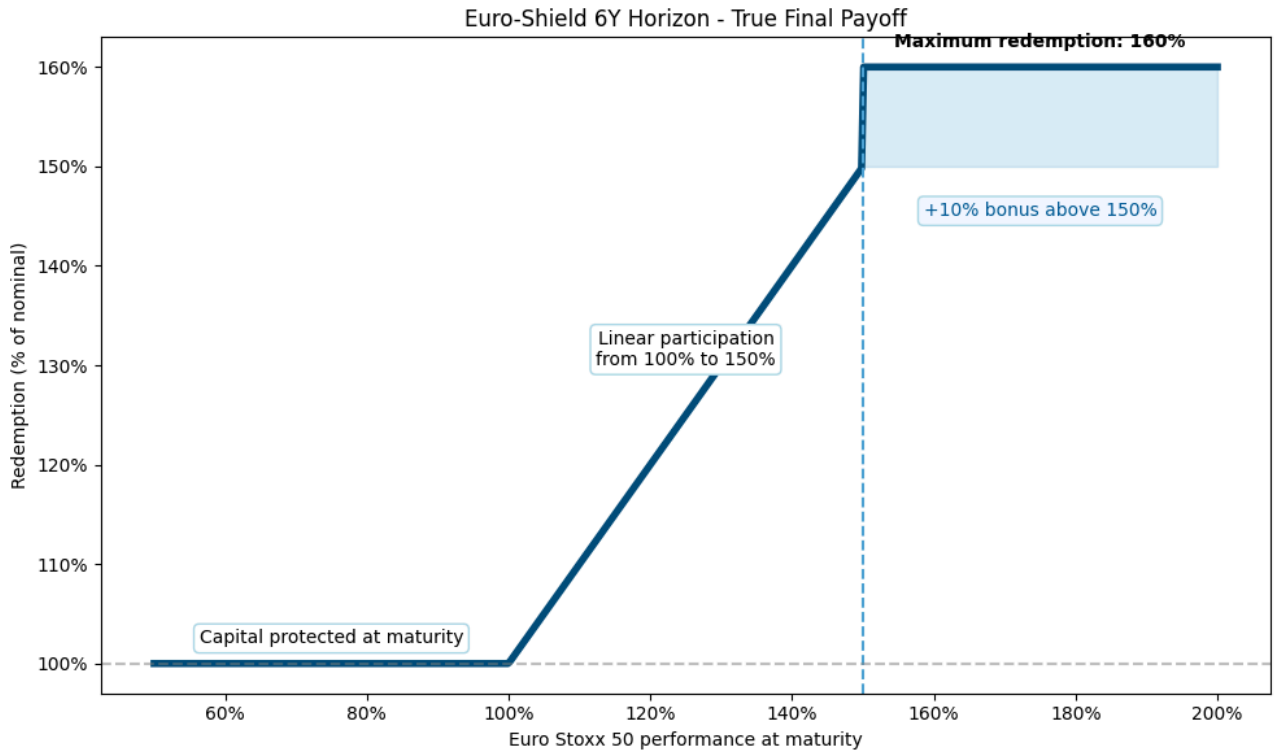
The product is engineered using three specific building blocks:

Block	Instrument	Quantity / Details
Lego 1	Zero-Coupon Bond	Face = 100€, T=6
Lego 2	Long ATM Call	$K = 3,600$, Qty ≈ 0.0278
Lego 3	Short OTM Call	$K = 5,760$ (160%), Qty ≈ 0.0278

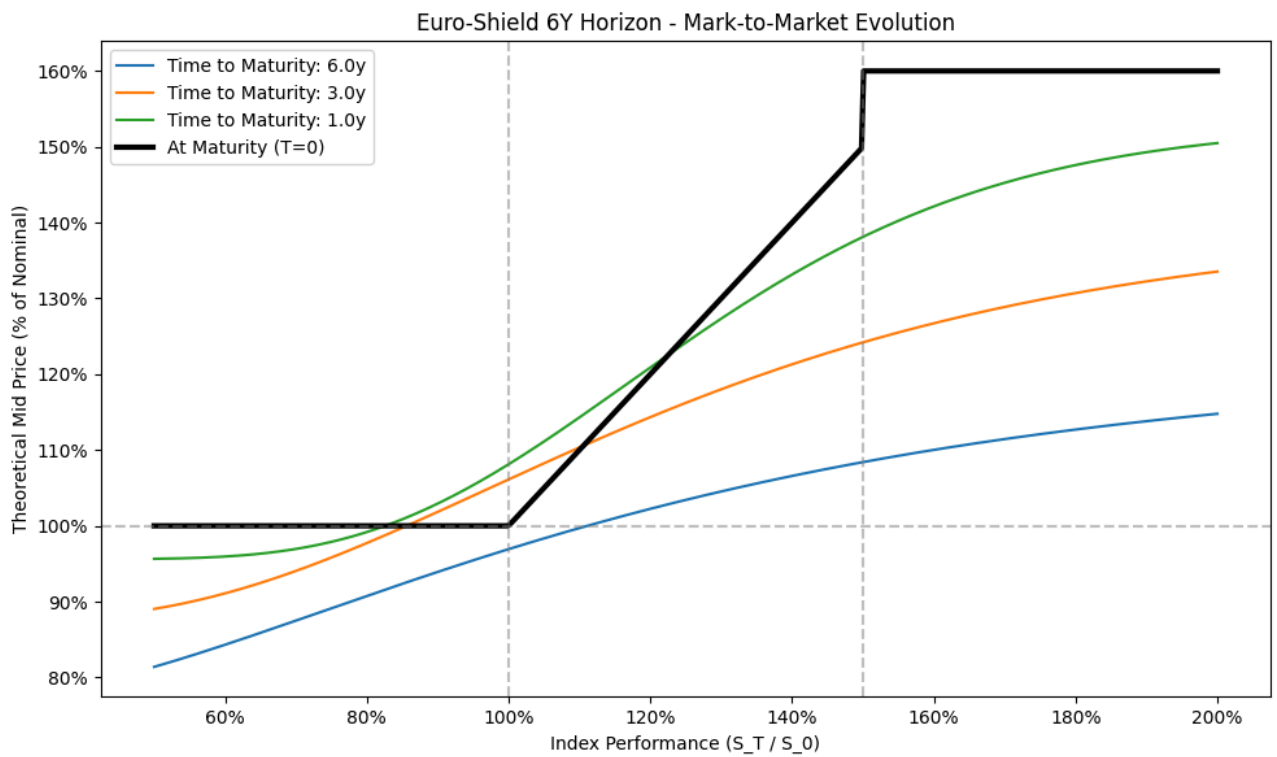
Lego Block Visualization



Payoff Diagram at Maturity



Mark-to-Market Evolution



Step 2: Yield Curve Bootstrapping

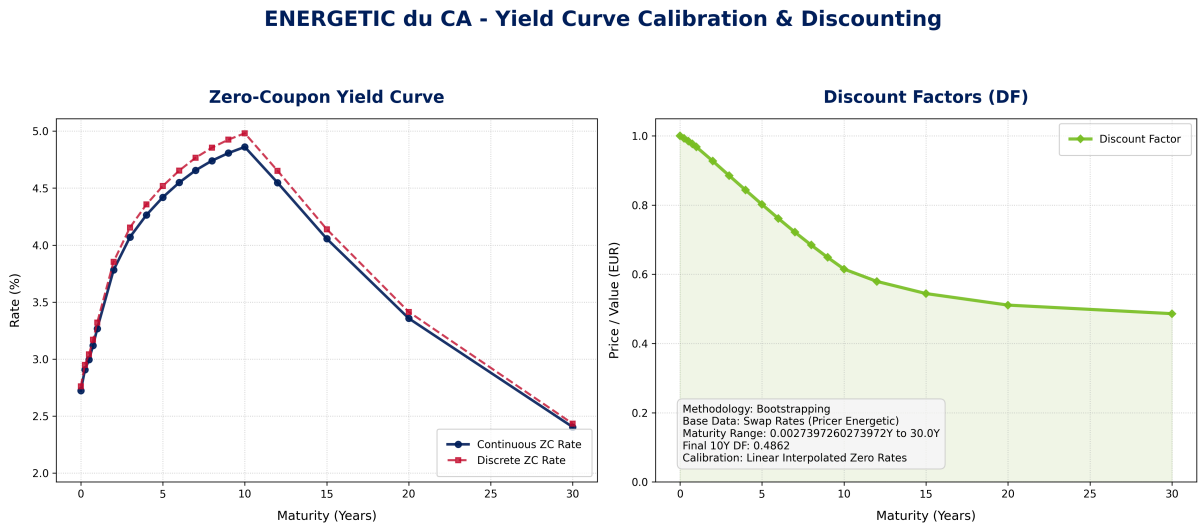
Logic: Extract Zero-Coupon rates from market Swap rates.

- **Formula (Integer T):**

$$\mathrm{DF}_T = \frac{1 - \mathrm{Rate}_T \sum_{\{i=1\}}^{\{T-1\}} \mathrm{DF}_i}{1 + \mathrm{Rate}_T}$$

- **Formula (Continuous r):**

$$\mathrm{ZC}_{\mathrm{cont}} = -\frac{\ln(\mathrm{DF}_T)}{T}$$



Bootstrapping Results

Critical data extracted for pricing models:

Maturity	Swap Rate	Discount Factor	ZC Cont (r_c)
1 Year	3.320%	0.9678	3.266%
5 Years	4.480%	0.8017	4.383%
6 Years (T)	4.605%	0.7612	**4.502%**
7 Years	4.708%	0.7218	4.656%

Step 3: Black-Scholes Pricing

- **Spot (S_0):** 3,600 €
- **Risk-free Rate (6Y):** 4.502%
- **Volatility ATM (K=3600):** 27.53%
- **Volatility Capped (K=5400):** 24.00%

Component	Value (€)	Notes
ZC Bond (Lego 1)	76.33 €	Capital guarantee
ATM Call (Lego 2)	1,331.51 €	Per unit of index
Cap (Lego 3)	564.68 €	Short option premium

Step 4: Bank Margin Calculation

- **Lego 1 (ZC Bond):** $100 \times 0.7612 = 76.33$ €
- **Option Spread Cost:** $(1,331.51 - 564.68) \times 0.0278 = 20.98$ €

- **Manufacturing Cost:** $76.33 + 20.98 =$ **97.31 €**
- **Selling Price: 100.00 €**
- **Upfront Bank Margin: 2.69%**

Step 5: Greeks Analysis (Day 1)

Exposure per 100€ nominal sold:

Greek	Exposure	Risk Interpretation
Delta (Δ)	1. 0.0081	Long Delta — Trader must hedge by selling index.
Vega (ν)	€ - 0.2337	Short Vega — Capped option dominates sensitivity.
Rho (ρ)	€ - 0.0409	Short Rho — Rates rise decrease bond value.
Theta (Θ)	1. 0.0091	Positive Theta — Accrued interest > time decay.

Step 6: Sensitivity Analysis

Impact of market movements on the 2.69% baseline margin.

Scenario	New Margin	Interpretation
Rates Fall (-50bp)	0.62%	High risk — ZC Bond cost surges.
Vol Rises (+1%)	2.93%	Benefit — Product is Short Vega.
Extend to 7Y	6.81%	Benefit — High discount on ZC.
Cap Tightened (59%)	2.78%	Benefit — More premium sold.

Step 7: Secondary Market Valuation

Assume 1 year has passed ($T = 5$ remaining), Spot is unchanged.

Instrument	**Value (T=1)**	**Change vs T=0**
ZC Bond (Lego 1)	80.32 €	1. 4.06 € (Accrual)
Options (Lego 2 & 3)	20.96 €	• 0.07 € (Decay)
Total Mid-Price	**101.28 €**	**+ 3.99 €**

Key takeaway: The product gains value even if the market is flat, confirming positive Theta accrual.

Euro-Shield 6Y Horizon | Structured Product

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